

Problem 1

If $\phi : G \rightarrow H$ is an isomorphism, prove that $|\phi(x)| = |x|$ for all $x \in G$. Deduce that G and H have the same number of elements of order n for each $n \in \mathbb{Z}^+$. Is this result true if ϕ is only a homomorphism?

Solution:

Let x have order $n \Rightarrow x^n = 1_G$

Let $\phi(x)$ have order $m \Rightarrow (\phi(x))^m = 1_H$

Firstly, consider that $(\phi(x))^m = \phi(x^m) = 1_H \Rightarrow x^m = 1_G$ since $\phi(1_G) = 1_H$

Then $|x| \leq m$ i.e. $|x| \leq |\phi(x)|$

Next, consider that $(\phi(x))^n = \phi(x^n) = \phi(1_G) = 1_H$

Then $|\phi(x)| \leq n$ i.e. $|\phi(x)| \leq |x|$

Then the claim follows

Since $\phi(x)$ is bijective and preserves the order of any $x \in G$ then we can deduce that G, H have the same number of elements of order $n \in \mathbb{Z}^+$. If there was some x such that $|x| \neq |\phi(x)|$ then there exists some other $y \in G, x \neq y$ that has order $\phi(x)$, but y should have order $\phi(y)$ which is a contradiction in general

If ϕ is a homomorphism then the result may not hold, for example in the case of the trivial homomorphism

Problem 2

1. Prove that the multiplicative $\mathbb{C}^*$ and $\mathbb{R}^*$ are not isomorphic.

2. Prove that $\mathbb{Z}$ and $\mathbb{Q}$ are not isomorphic

Solution:

1. Consider $z = e^{\frac{2\pi i}{3}} \in \mathbb{C}$ which has order 3, but there are no positive real numbers which have order 3 (only 1, -1 in $\mathbb{R}$ have finite order with orders 1 and 2)

Using problem 1 the claim follows

2. Suppose that $\varphi : \mathbb{Q} \rightarrow \mathbb{Z}$ is an isomorphism. Then for some $q \in \mathbb{Q}$ we have $\varphi(q) = 1 \in \mathbb{Z}$

Then

$$1 = \varphi(q) = \varphi\left(2\frac{q}{2}\right) = \varphi\left(\frac{q}{2} + \frac{q}{2}\right) = \varphi\left(\frac{q}{2}\right) + \varphi\left(\frac{q}{2}\right) = 2\varphi\left(\frac{q}{2}\right)$$

But there is no $z \in \mathbb{Z}$ such that $2z = 1$ and the claim follows

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Problem 3

Let G be a group.

1. Prove that the map from G to itself defined by $g \mapsto g^{-1}$ is an automorphism iff G is Abelian
2. Prove that the map from G to itself defined by $g \mapsto g^2$ is a homomorphism iff G is abelian

Solution:

Denote the maps as $\varphi : G \rightarrow G$

1.

($\Leftarrow$) suppose that G is abelian. Then φ is a homomorphism since for any $g_1, g_2 \in G$ we have $\varphi(g_1 g_2) = (g_1 g_2)^{-1} = g_2^{-1} g_1^{-1} = g_1^{-1} g_2^{-1} = \varphi(g_1) \varphi(g_2)$

φ is bijective since for every $g \in G$ we have that the preimage of g under φ is g^{-1} (inverses exist) so that φ is surjective and that $\varphi(g_1) = \varphi(g_2) \Leftrightarrow g_1^{-1} = g_2^{-1} \Rightarrow g_1 = g_2$ so that φ is injective

($\Rightarrow$) suppose that φ is an automorphism. Let $x = \varphi(a)$ and $y = \varphi(b)$ where $x, y, a, b \in G$. a, b must exist since φ is a bijection. Then $xy = \varphi(a) \varphi(b) = \varphi(ab) = (ab)^{-1} = b^{-1} a^{-1} = \varphi(b) \varphi(a) = yx$ so that G is abelian

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2.

($\Leftarrow$) suppose that G is abelian. Then for any $g_1, g_2 \in G$ we have $\varphi(g_1 g_2) = (g_1 g_2)^2 = g_1^2 g_2^2 = \varphi(g_1) \varphi(g_2)$ where the last equality holds due to the assumption

($\Rightarrow$) suppose that φ is a homomorphism. then we must have for any $g_1, g_2 \in G$ that $\varphi(g_1 g_2) = \varphi(g_1) \varphi(g_2)$

This means that $(g_1 g_2)^2 = g_1^2 g_2^2$ is true, which implies that G is abelian (from group properties we learned previously)

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Problem 4

Prove that the map $A \mapsto (A^T)^{-1}$ is an automorphism on $GL_n(\mathbb{R})$

Solution:

Denote the map by $\varphi : GL_n(\mathbb{R}) \rightarrow GL_n(\mathbb{R})$

Let $A_1, A_2 \in GL_n(\mathbb{R})$

We can use the fact that $(A_1 A_2)^T = A_2^T A_1^T$ and that $(A^T)^T = A$

First we can show that φ is a homomorphism:

$$\varphi(A_1 A_2) = ((A_1 A_2)^T)^{-1} = (A_2^T A_1^T)^{-1} = (A_1^T)^{-1} (A_2^T)^{-1} = \varphi(A_1) \varphi(A_2)$$

Next, we can show φ is a bijection:

φ is surjective since if we have an element $A \in GL_n(\mathbb{R})$ then its preimage is $(A^{-1})^T$ since applying φ gives $\varphi((A^{-1})^T) = \left(\left((A^{-1})^T \right)^T \right)^{-1} = A$ by the above facts

φ is injective since if $\varphi(A_1) = \varphi(A_2) \Leftrightarrow (A_1^T)^{-1} = (A_2^T)^{-1} \Leftrightarrow A_1^T = A_2^T \Leftrightarrow A_1 = A_2$

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References:

https://proofwiki.org/wiki/Transpose_of_Matrix_Product

Problem 5

Let G be a cyclic group. Find all automorphisms of G . Consider both the finite order and the infinite order case

Solution:

Let $n = |G|$

1. If G is cyclic and finite order n then the automorphisms of G are the set

$$\text{Aut}(G) = \{\varphi_m : \varphi(x) = x^m : 1 \leq m < n \wedge (m, n) = 1, \forall x \in G\}$$

where $\varphi : G \rightarrow G$

Note that $\forall x, y$ that $\varphi(xy) = (xy)^m = x^m y^m = \varphi(x)\varphi(y)$ since G is cyclic and therefore abelian

Note that if $\varphi(x) = \varphi(y)$ then $x^m = y^m \Rightarrow x = y$ since G is cyclic and $(m, n) = 1$ and by Bezout identity we can write $1 = am + bn$ for some a, b and then

$$x^m = y^m \Leftrightarrow (x^m)^a = (y^m)^a \Leftrightarrow x^{am} = y^{am} \Leftrightarrow (x)^{1-bn} = y^{1-bn} \Leftrightarrow x(x^n)^{-b} = y(y^n)^{-b} \Leftrightarrow xe = ye \Leftrightarrow x = y$$

Combined with the fact that φ maps a finite set to itself, then φ is surjective and bijective, as well as an automorphism

If we let g be a generator of G then g^m is a generator as well (due to the coprime condition) however if $n = bm$ for some integer b then we would have $\varphi(1) = 1$ and

$$\varphi(g^m) = g^{(m)\frac{n}{b}} = g^{n\frac{m}{b}} = 1$$

which violates bijectivity. Therefore the condition that $(m, n) = 1$ must hold (φ can only be of the form described above) and $|\text{Aut}(G)| = \phi(n)$

This means also that φ must map a generator to another generator if φ is to be an automorphism of a cyclic (finite) group

2. If G is infinite order then the automorphisms of G are the identity mapping and the inverse mapping

Since inverses are unique, then the inverse map $\varphi : G \rightarrow G$ s.t. $\varphi(x) = x^{-1} \forall x \in G$ is bijective and $\varphi(xy) = (xy)^{-1} = y^{-1}x^{-1} = x^{-1}y^{-1} = \varphi(x)\varphi(y)$ due to G is abelian

The identity mapping here is also an automorphism

From class we have that G is isomorphic to $\mathbb{Z}$ (see the reference)

References:

<https://www.youtube.com/watch?v=bNJDQWNwb8o>

Jose Carlos Santos: <https://math.stackexchange.com/questions/4086992/why-does-the-set-of-all-automorphisms-on-the-additive-group-of-integers-contain>

Problem 6

Prove that the homomorphism $S_3 \rightarrow \text{Aut}(S_3)$ defined by $g \mapsto \phi_g$ is an isomorphism. Here ϕ_g denotes the 'conjugation by g ' function defined in class.

Solution:

denote this homomorphism by $\varphi : S_3 \rightarrow \text{Aut}(S_3)$

for any $g \in G$, conjugation by g is the map $\phi_g : G \rightarrow G$ defined by $\phi_g(x) = gxg^{-1} \forall x \in G$

First we show that φ is a homomorphism (let $g_1, g_2 \in S_3$):

note that $\varphi(g_1 g_2) = \phi_{g_1 g_2}$ and that $\varphi(g_1) \varphi(g_2) = \phi_{g_1} \phi_{g_2}$ so we want to show that $\phi_{g_1 g_2} = \phi_{g_1} \phi_{g_2}$

let $\sigma \in S_3$ then $\phi_{g_1 g_2}(\sigma) = g_1 g_2 \sigma (g_1 g_2)^{-1} = g_1 g_2 \sigma g_2^{-1} g_1^{-1}$

and that $\phi_{g_1}(\phi_{g_2}(\sigma)) = \phi_{g_1}(g_2 \sigma g_2^{-1}) = g_1 g_2 \sigma g_2^{-1} g_1^{-1}$ so they match, therefore φ is a homomorphism

Next we show that φ is bijective:

First we can show injective:

note $\varphi(g_1) = \varphi(g_2) \Leftrightarrow \phi_{g_1} = \phi_{g_2}$ and note that for any $\sigma \in S_3$ that $\phi_{g_1}(\sigma) = g_1 \sigma g_1^{-1}$ and $\phi_{g_2}(\sigma) = g_2 \sigma g_2^{-1}$

we want to show that $g_1 \sigma g_1^{-1} = g_2 \sigma g_2^{-1} \Rightarrow g_2^{-1} g_1 \sigma g_1^{-1} g_2 = \sigma \Rightarrow (g_2^{-1} g_1) \in Z(S_3) = \{1\}$

then $g_2^{-1} g_1 = 1 \therefore g_2 = g_1$ and φ is injective

next we can show surjective by cardinality argument:

$|S_3| = 3! = 6$ and we want to show that $|\text{Aut}(S_3)| = 6$.

Since S_3 has 6 bijections then $\text{Aut}(S_3)$ must have at most 6 bijections (since bijections in $\text{Aut}(S_3)$ must also be homomorphisms) and due to injectivity we have that $|\text{Aut}(S_3)| = 6$

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Problem 7

Let H be the cyclic subgroup of the alternating group A_4 generated by the permutation (123) . Exhibit the left and right cosets of H explicitly.

Solution:

the alternating group $A_4 =$

$\{(), (12)(34), (13)(24), (14)(23), (123), (132), (124), (142), (134), (143), (234), (243)\}$

and $H = \langle (123) \rangle = \{(), (123), (132)\}$

The number of cosets are $|A_4|/|H| = 4$ by a proposition in class (the left/right cosets also form a partition of $G = A_4$)

Then the left cosets of H are:

handwritten

And the right cosets of H are:

handwritten

Problem 8

Prove that if H and K finite subgroups of G whose orders are relatively prime, then $H \cap K = 1$

Solution:

Firstly, we can try to show that $H \cap K \leq H \wedge H \cap K \leq K$. Since $1 \in H \wedge 1 \in K$ then $H \cap K \neq \emptyset$.

Additionally, $\forall x, y \in H \cap K$ we have that $x, y \in H \wedge x, y \in K \Rightarrow y^{-1} \in H \wedge y^{-1} \in K$ and so $xy^{-1} \in H \wedge xy^{-1} \in K \Rightarrow xy^{-1} \in H \cap K$ due to closure and existence of inverse. So this is true.

Then by Langrange's theorem we can say that $|H \cap K|$ should divide both $|H|$ and $|K|$

Since $(|H|, |K|) = 1$ then $|H \cap K| = 1 \Rightarrow H \cap K = \{1\}$

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Problem 9

Prove that if G is a group of prime order p , then G is a cyclic group.

Solution:

Note that every element of G generates a cyclic subgroup of G

Consider the finite subgroup $\langle a \rangle \leq G$ for some $a \in G$

If $G = \{1\}$ then $a = 1$ and here G is cyclic

Otherwise, let $a \neq 1$ then $|\langle a \rangle|$ divides $|G|$ by Langrange's theorem and since $|G|$ is prime then $|\langle a \rangle| = p = |G|$ implying that $G = \langle a \rangle$, since a generates a subgroup of G with order equal to G thereby generating all of G

Then G is cyclic.

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references:

<https://math.stackexchange.com/q/3293910>